A Thirteen-Integer Jones Equation with Degree-Separated Coding, CRT, and Coprime Guards

Mathematical research artifact for ProveIt

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1 Thirteen integers: CRT elimination and a coprime spectator guard

The polynomial specified below has exactly thirteen unrestricted integer unknowns, exact total degree 11 546 112, and a division-free circuit of 520 live binary operations (328 multiplications, 169 additions and 23 subtractions). Its integer zero existence is equivalent to a finite syntactic ZFC refutation. Thus, with the stated inherited compiler theorem, ZFC proves that its lack of integer zeros is equivalent to Con(ZFC). This is an ordinary mathematical proof, not a new Lean certificate, a proof of consistency, or a global minimality claim. Unlike the ten-integer companion, the arithmetic finalizer here uses only the established Jones ratio interface, not Sun's 2026 bridge. The recommended compact companion in Section 1.8 has degree 11 543 632 and 518 operations, with the same thirteen unknowns.

1.1 Degree-separated kernels and the complete polynomial

Use the original package-31 system of 22 quadratic residuals $r_j(p)$ in 31 strictly positive coordinates. The exact data are `split/parent_source.json`; the coherent fixed port tuple is $(\beta, K_0, M_C, M_F) = (\beta, K, M_C, M_{F,\mathrm{source}})$ from `FIXED_BINDING.json`. All four compressed little-endian payloads are included and hash-authenticated. The inherited compiler theorem identifies its positive common zeros with finite syntactic ZFC refutations. This is the original tuple, not a mixture with a later packet's smaller coefficients.

Perform just two elementary integer row operations:

$$\varrho_{14} = r_{14} + r_2, \qquad \varrho_{15} = r_{15} + r_3, \qquad \varrho_j = r_j \quad (j \neq 14, 15).$$

The inverses are subtraction of rows 2 and 3, so the common zero set is unchanged. Write $\varrho_j = \sum_\alpha r_{j,\alpha} p^\alpha$ and

$$c_{j,\alpha} = \begin{cases} 2r_{j,\alpha}, & |\alpha| = 0, 1, \\ (\prod_i \alpha_i!) r_{j,\alpha}, & |\alpha| = 2. \end{cases}$$

Use the 31 positive weights w_i and the 22 targets q_j in `split/split_code.json`, and set

$$D_d(T) = \sum_{j,\, |\alpha| = d} c_{j,\alpha} T^{q_j - \alpha \cdot w} \quad (d = 0, 1, 2).$$

The supplied exact certificate establishes

$$W = \max_i w_i = 116, \quad Q_* = \max_j q_j = 2129, \quad (\deg D_0, \deg D_1, \deg D_2) = (1719, 2027, 1929).$$

The leading coefficient of D_2 is exactly 1; every kernel has internally distinct exponents. There are 89 source records. Checking all 22,544 same-degree source/expansion interactions proves

$$[T^{q_j}]\big(D_2(T)U(T)^2 + D_1(T)U(T) + D_0(T)\big) = 2\varrho_j(p), \qquad U(T) = \sum_i p_i T^{w_i}.$$

The degrees are kept in separate products. Thus a linear label may equal a quadratic label without causing cross-degree contamination. No homogeneous dummy coordinate or full Sidon condition is needed. Let

$$K_* = \max_d(\deg D_d + dW) = 2161, \qquad E_* = K_* + W + 2 = 2279.$$

The fixed coefficient bound, checked from the actual original port bit lengths, is $|c_{j,\alpha}| < A_* = 2^{2134927}$.

The unknowns are exactly

$$(\epsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, r, v) \in \mathbb{Z}^{13}.$$

All following expressions are polynomial abbreviations. Fix

$$\kappa = 2134940, \qquad a = 2^\kappa = 8192A_*, \qquad \eta = 2^{17079510} = 2^{8\kappa-10},$$

and define

$$\begin{aligned}
b &= 2\epsilon^2, & B &= ab^2,\\
\ell &= \sum_i B^{w_i}, & T_* &= \sum_j B^{q_j},\\
U_c &= \ell + g, & \mathcal{G} &= D_2(B)U_c^2 + D_1(B)U_c + D_0(B),\\
\mathcal{G}_+ &= \mathcal{G} + B^{2162}, & N_0 &= B^{117},\\
L &= 2bB^{2279}, & N &= \eta bB^{1201},\\
c &= bB^{116}, & G_c &= 32c,\\
M_c &= N_0 - 1 - (b-1)\ell, & J_c &= 2\mathcal{G}_+ + T_*,\\
S_c &= g + N_0 J_c, & T_c &= M_c + N_0(B-2)T_*,\\
R &= L^2 + (S_c + T_c)L + (L - 1 - S_c).
\end{aligned}$$

The extra monomial B^{2162} lies above every target. It repairs positivity on arbitrary signed g without changing an extracted residual or adding a new unknown. Every exponent in these formulas is fixed and nonnegative.

Use the Jones registers with the exact CRT reparameterization of H below, and $\omega = w^2$:

$$\begin{aligned}
U &= N^2\omega, & Y &= N^2 s,\\
M &= RY, & P &= 2M^2U,\\
K &= R + 1 + h(P-1), & A &= M(U+1),\\
B_1 &= 2R + 1, & W_1 &= b\omega,\\
C &= B_1 + W_1 + \phi, & J &= 4A - 5, & Q &= 3W_1C - 2(W_1^2 - 1),\\
D &= (A^2-1)C^2 + 1, & E &= iJDC^2, & F &= (A^2-1)E^2 + 1,\\
G &= A + F(CD + 1 - A), & H &= C + F(B_1 + jC), & I &= (G^2-1)H^2 + 1,\\
a_0 &= (P^2-1)K^2 + 1, & a_1 &= DFI,\\
u &= J, & t &= Q.
\end{aligned}$$

Set

$$\delta = C - KY, \qquad \Delta = g(c-g)K^2 - c^2\delta^2, \qquad (d_1, d_2, d_3) = (3s-1, 3\phi+2, 3\Delta-1).$$

The new coprime-spectator stack is

$$C_0 = a_0, \qquad B_j^* = 1 + 4d_jC_{j-1}z_j^2, \qquad C_j = C_{j-1}B_j^* \quad (j = 1, 2),$$

$$B_3^* = 1 + 4d_3C_2a_1z_3^2, \qquad C_* = C_3 = C_2B_3^*.$$

Thus the last guard contains the second square candidate a_1 as a spectator. This is the only change to the Pell sign stack. Finally define

$$\begin{aligned} Z &= \epsilon wiz_1z_2z_3a_0, & \Sigma &= Zu^5, & \rho &= t + ur, \\ \Theta &= (\rho^3 - u^2\rho - u^3)(\rho^2 + 23u^2), & X &= \Sigma v - \Theta, \\ A_0 &= \Sigma^2 C_*, & A_1 &= \Sigma^2 a_1. \end{aligned}$$

The requested thirteen-integer equation is the following single polynomial:

$$\boxed{\mathcal{F}_{13} = (X^2 - A_0 - A_1)^2 - 4A_0A_1 = 0.} \tag{1}$$

The literal division-free DAG is `generated/signed_coprime_split_crt.json`.

1.2 Signed-input positivity and sharp block bounds

Assume only $\epsilon \neq 0$, hence $b \geq 2$ and $B = 8192A_*b^2$. Geometric-series estimates give

$$D_2(B) > \tfrac{1}{2}B^{1929}, \quad |D_1(B)| < 2A_*B^{2027}, \quad |D_0(B)| < 2A_*B^{1719}.$$

Completing half of the quadratic square gives, for every real U_c,

$$\begin{aligned} \mathcal{G}_+ &\geq \tfrac{1}{2}D_2(B)U_c^2 + B^{2162} - \frac{D_1(B)^2}{2D_2(B)} - |D_0(B)| \\ &> \tfrac{1}{4}B^{1929}U_c^2 + B^{2162} - 4A_*^2B^{2125} - 2A_*B^{1719} \\ &> \tfrac{1}{4}B^{1929}U_c^2 + \tfrac{1}{2}B^{2162}. \end{aligned}$$

The last line uses $4A_*^2/B^{37} < 1/4$ and $2A_*/B^{443} < 1/4$. For every *integer* g, $U_c = \ell + g$ is an integer. Therefore

$$S_c > \tfrac{1}{2}N_0B^{1929}U_c^2 + U_c + N_0B^{2162} - \ell > 0.$$

Indeed $tU_c^2 + U_c \geq 0$ for an integer U_c and $t \geq 1$, while $\ell < N_0$. This explicitly absorbs the signed linear term g, rather than incorrectly deducing $S_c > 0$ merely from $J_c > 0$.

The mask satisfies $0 < M_c < N_0$, so $T_c > 0$ as well. Consequently

$$R = L^2 + S_c(L-1) + T_cL + L - 1 > L^2.$$

Also

$$N^2 = 2^{-20}a^{15}B^{2403} < B^{2418} < B^{4558} < L^2, \qquad b \leq c \leq N.$$

Thus $R \geq N \geq 8$, $b \leq R, N$, and all these estimates hold before any sign guard, carry condition, or source-coordinate decoding.

For the range needed by the interval guard, assume $0 < g < c$. Since $\ell < 2B^{116}$, one has $U_c < 2bB^{116}$. Then

$$|\mathcal{G}| < 12A_*b^2B^{2161}, \qquad 2|\mathcal{G}| < B^{2162}/256, \qquad T_* < B^{2162}/256.$$

Writing $P_c = B^{2162}$, this gives $0 < J_c < (2 + 1/128)P_c$. Since $g < N_0$ and $P_c > 128$,

$$0 < S_c < 3B^{2279}, \qquad 0 < T_c < B^{2279}, \qquad S_c + T_c < 4B^{2279} \leq L.$$

For the middle bound, use $(B-2)T_* < B^{2130}$ and $1 + B^{2130} < B^{2162}$. These sharp estimates are why the off-target padding does not require increasing the block exponent from 2279 to 2280.

1.3 Sparse carries and parity-tuned square threshold

Continue to assume $0 < g < c$, and suppose now $b = 2^k$, so $B = 2^\nu$ with $\nu = \kappa + 2k$. The low mask gives

$$g \,\&\, M_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B^{w_i}, \qquad 0 \le n_i < b.$$

Here the binary "and" is proof notation only. Put $p_i = n_i + 1$, so $U_c = \sum_i p_i B^{w_i}$. Every coefficient h_e of $D_2(T)U(T)^2 + D_1(T)U(T) + D_0(T) + T^{2162}$ obeys

$$|h_e| < 994 A_* b^2 < B/8.$$

The bound includes the possible collisions at unmarked coefficients and the extra coefficient 1; it follows from $\sum_i p_i \le 31b$. The unnormalized digits of J_c are $2h_e + \varepsilon_e$, where $\varepsilon_e = 1$ exactly at the targets. Starting with incoming carry 0, normalized carries remain in $\{-1, 0\}$. At a target, its digit avoids $B - 2$ exactly when it is 0 or 1. This happens exactly when $h_e = 0$, independently of an incoming borrow. The degree-separated extraction theorem then gives, using the inverse row operations,

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad p = n + 1 \text{ is a positive common zero of the original system.}$$

Conversely every such zero has this code with $0 < g < c$ for arbitrarily large $b = 2^{1+2r}$. The all-one positive tuple is not a zero, since original row 5 has value 1 there. Thus the encoded nonnegative tuple is not zero.

Write $L = 2^\lambda$. Since $S_c, T_c > 0$ and $S_c + T_c < L$, the three nonoverlapping blocks in R give

$$s_2(R) = \lambda + s_2(T_c) + 1 - \tau_2(S_c, T_c),$$

where $\tau_2(S, T) = s_2(S) + s_2(T) - s_2(S + T) \ge 0$ is the number of binary carries. The 31 low allowed positions and 22 high target positions give

$$s_2(T_c) = 139\nu - 31k - 22, \qquad \lambda = 1 + k + 2279\nu.$$

Therefore

$$\begin{aligned} \lambda + s_2(T_c) + 1 &= 2418\kappa + 4806k - 20 \\ &= 2\big((8\kappa - 10) + k + 1201(\kappa + 2k)\big) = \log_2 N^2. \end{aligned}$$

Legendre's identity $v_2\binom{2R}{R} = s_2(R)$ now proves

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad S_c \,\&\, T_c = 0.$$

The factor b in N is necessary in this parity class. It is a polynomial factor, not a new witness or a variable exponent.

1.4 An exact polynomial CRT reparameterization

The inherited ratio interface uses $H_{\text{old}} = B_1 + j_{\text{old}} C$ and $JF \mid QF + J(H_{\text{old}} - C)$. Since $E = iJDC^2$,

$$F \equiv 1 \pmod{J}, \qquad F \equiv 1 \pmod{C},$$

and hence $\gcd(J, F) = 1$. The divisibility is equivalent to $J \mid Q$ and $F \mid H_{\text{old}} - C$. Our changed register

$$H = C + F(B_1 + jC)$$

automatically satisfies the latter. It also has the original form, with

$$j_{\mathrm{old}} = 1 + (A^2 - 1)i^2J^2D^2C^3B_1 + Fj,$$

which is a polynomial integer, since $F - 1 = (A^2 - 1)i^2J^2D^2C^4$. This identity holds for arbitrary signed inputs, including $C = 0$.

Conversely, at the canonical positive ratio witnesses, $0 < B_1 < C$ and $j_{\mathrm{old}} \geq 1$. Therefore $z = (H_{\mathrm{old}} - C)/F > 0$ and $z \equiv B_1 \pmod{C}$, so $z = B_1 + jC$ with $j \geq 0$. Thus every canonical witness is represented. No extra unknown or cofinal restriction is introduced. The merger now has $u = J$ rather than JF. This CRT step is inherited from the separately identified sparse-Pell packet, not claimed as an original theorem here.

1.5 The interval guard and the Jones interface

If $\Delta > 0$ and $b \geq 2$, then $K \neq 0$ and $0 < g < c$. The identity $4g(c - g) = c^2 - (c - 2g)^2 \leq c^2$ implies $4\delta^2 < K^2$. Along with $s \geq 1$, $\phi \geq 0$, and $w \neq 0$, this gives $C, Y > 0$ and $K > 0$. The usual square-gap argument forces $h \geq 1$: if $h \leq 0$, then $1 \leq K \leq R + 1 < P$; $K = 1$ would force $P - 1 \mid R$, which is impossible, while for $2 \leq K < P$,

$$(PK - 1)^2 < (P^2 - 1)K^2 + 1 < (PK)^2.$$

Thus a_0 cannot be a square in that branch.

For precision, instantiate the free ceiling in `Jones1982.Nine.ratio_sound` by $G_{\mathrm{aux}} = 8gK^2+1$, rather than assuming its old margin condition. The positive integer $K^2 - 4\delta^2$ then gives

$$(K^2 - 4\delta^2)G_{\mathrm{aux}} - 8gK^2 > 0.$$

This ceiling is a proof-side abbreviation; the ratio registers do not otherwise depend on it.

Replace i by $|i|$, which changes no condition. To handle signed j, $u \mid t$ now means $J \mid Q$, while $F \mid H - C$ holds identically. Also $F \equiv 1 \pmod{D}$, $G \equiv 1 \pmod{D}$, and $G \equiv A \pmod{F}$ show that the positive integers D, F, I are pairwise coprime. Hence $a_1 = DFI$ square makes each a square. Put $d = G^2 - 1 > 0$ and $x = \sqrt{I}$. The norm-one unit $x + H\sqrt{d}$ is positive even if $H < 0$. A positive power of $G + \sqrt{d}$ is the identity modulo CF; multiplying by sufficiently large such powers gives a Pell solution $H' > B_1$ with $H' \equiv H \pmod{CF}$. Then $j' = (H' - B_1)/C > 0$, all square conditions remain true, and $J \mid Q$ remains true. The literal positive Jones interface therefore applies and gives the power-of-two base and central-binomial condition required above.

For completeness, take any common source zero, choose $b = 2^{1+2r}$ large enough, and set $\epsilon = 2^r$. The direct source code has $32g < G_c$, so the literal ratio completeness theorem supplies its canonical witnesses. In `Ratio.lean`, `exists_B25core` sets

$$\omega = 2^{2R+1-k}, \qquad U = (N^2/b)2^{2R+1}.$$

Since $k = 1 + 2r$ is odd and $k < 2R + 1$, this ω is a positive square. No claim that the canonical s is a square is needed.

The same canonical construction has $\rho_0 = (U + 1)^{2R}/U^R = Y + f$ with $0 \leq f < 2^{2R}/U = b/(2N^2)$ and $Y \geq 1$. The source's `ratio_core` gives

$$|C/K - \rho_0| \leq \frac{2(Y + f)}{Y(U + 1)} < \frac{4}{U}, \qquad |C/K - Y| < \frac{b}{N^2}.$$

The last inequality uses $R \geq 8$. Since $c, b \leq N$ and $g(c - g) \geq 1$, it follows that

$$\frac{c^2(C - KY)^2}{K^2} < \frac{c^2b^2}{N^4} \leq 1 \leq g(c - g),$$

so the very same canonical witnesses, translated by the exact CRT formula, satisfy $\Delta > 0$. This is a cofinal square-base proof, not an unsupported restriction of an existing slack variable to squares.

1.6 Raw two-root recovery, including the zero spectator branch

For every integer assignment, $u = J = 4A - 5 \neq 0$. The polynomial

$$\Pi(x) = (x^3 - x - 1)(x^2 + 23)$$

has no rational root and has a root modulo every positive integer. For an odd prime other than 23, either the cubic has a simple root or it is irreducible, whose Frobenius three-cycle makes its nonzero discriminant -23 a square. The quadratic then has a simple root. At 23 the cubic has the simple root 3; at 2, $-23 \equiv 1 \pmod 8$ admits compatible square roots. Hensel lifting and CRT finish the assertion. Thus $\Theta = u^5\Pi(t/u + r) \neq 0$ for every assignment, and

$$\exists r \quad Zu^5 \mid \Theta \quad \Longleftrightarrow \quad Z \neq 0 \text{ and } u \mid t.$$

Indeed a root modulo $|Z|$ proves the reverse implication once $u \mid t$. If $v_p(u) = \alpha > \beta = v_p(t)$, the two factors of Θ have valuations 3β and 2β, contrary to $u^5 \mid \Theta$. If $Z = 0$, divisibility would require $\Theta = 0$, already impossible.

At a zero of the equation, $\Sigma = 0$ is impossible because then the polynomial is $\Theta^4 \neq 0$. In particular $\epsilon, w, i, z_1, z_2, z_3, a_0$ are nonzero. Every $d_j \equiv 2 \pmod 3$ is nonzero, and every $B_j^* \equiv 1 \pmod 4$ is nonzero. Thus C_2, C_* are nonzero.

No coincident radicands. One has $C_* \neq a_1$ without any positivity assumption. If $a_1 = 0$ this follows from $C_* \neq 0$. Otherwise $B_3^* \neq 1$, and it cannot be -1 because it is 1 modulo 4. But

$$\gcd(B_3^*, a_1) = 1,$$

while $C_* = a_1$ would make B_3^* divide a_1. This would force $B_3^* = \pm 1$, a contradiction.

Put $q = X/\Sigma$. The raw norm equation implies

$$(q^2 + C_* - a_1)^2 = 4C_*q^2.$$

If $q = 0$, the norm would give $C_* = a_1$, just excluded. Therefore

$$r_0 = \frac{q^2 + C_* - a_1}{2q}, \qquad r_1 = q - r_0$$

are rational, with $r_0^2 = C_*$ and $r_1^2 = a_1$. Rational square roots of integers are integers. Hence C_*, a_1 are integer squares, $q \in \mathbb{Z}$, and $\Sigma \mid \Theta$. In particular $J \mid Q$. No complex-root separation assumption is used.

Recover the first two guards before ruling out $a_1 = 0$. The factors a_0, B_1^*, B_2^*, B_3^* are pairwise coprime, because each new factor is 1 modulo its entire preceding product. Their nonzero product C_* is a square. Thus each absolute factor is a square. Since $B_j^* \equiv 1 \pmod 4$, every B_j^* is positive: a negative square cannot be 1 modulo 4. The product sign then forces $a_0 > 0$, also a square. All prefixes are now positive, so $d_1, d_2 > 0$, yielding $s \geq 1$ and $\phi \geq 0$.

At this point the last factor might still equal 1 if $a_1 = 0$; this branch must not be silently discarded. The unconditional source estimate above has already proved $R > L^2 > 0$ for *every* signed g. Since $\epsilon, w \neq 0$ and $s \geq 1$, we have $b \geq 2$, $N \geq 8$, $U, Y > 0$, $A \geq 2$, and, using $\phi \geq 0$, $C > 0$. Then

$$D \geq A^2 > 0, \qquad F \geq 1, \qquad CD + 1 - A > 0, \qquad G \geq A > 1, \qquad I \geq 1.$$

Consequently $a_1 = DFI > 0$. Now $B_3^* \neq 1$, and its positivity with $C_2 a_1 z_3^2 > 0$ forces $d_3 > 0$, hence $\Delta > 0$. This closes the zero-spectator branch without adding a_1 to the nonzero multiplier. The interval and ratio arguments above therefore apply.

Completeness. Start from the canonical positive ratio witnesses, translated by the CRT parameterization. They have a_0, a_1 positive squares, $J \mid Q$, and all three strict sign conditions. For $j = 1, 2$, $d_j C_{j-1}$ is positive nonsquare because $d_j \equiv 2 \pmod 3$ and C_{j-1} is a square. A nontrivial Pell solution $x^2 - d_j C_{j-1} y^2 = 1$ can be chosen with y even, so $z_j = y/2 \neq 0$ makes $B_j^* = x^2$. For the last guard, $d_3 C_2 a_1$ is likewise positive nonsquare, since $C_2 a_1$ is a positive square. The same construction gives B_3^* square and C_* square. The intersective lemma chooses r with $\Sigma \mid \Theta$. Finally choose

$$v = \Theta/\Sigma + \sqrt{C_*} + \sqrt{a_1} \in \mathbb{Z}.$$

Then the raw norm vanishes. Both directions are proved. By the inherited compiler theorem, integer-zero existence is equivalent to a finite ZFC refutation, and ZFC proves its negation equivalent to Con(ZFC).

1.7 Exact degree, comparison, and verification scope

The exact relevant register degrees are

	a_0	a_1	C_2	C_*	Σ	X
deg	261250	2635110	1045009	4875914	448571	448572

and $\deg A_0 = 5773056$, $\deg A_1 = 3532252$. The degree of A_0 strictly exceeds those of X^2 and A_1, so the output has leading term A_0^2. Its diagonal leading coefficient is

$$116361\, a^{2885464} \eta^{960} \pmod{1000003},$$

which is nonzero since a and η are powers of two. The circuit upper bound and this specialization prove exact degree 11 546 112.

The previous seeded thirteen-integer construction has degree 57 866 184 and 759 operations. At its unchanged old source, the coprime spectator alone gives degree 46 680 720 and 758 operations, and adding CRT gives degree 28 847 344 and 754 operations. With degree-separated extraction and the one-monomial signed-input repair, the canonical result is degree 11 546 112 and 520 operations: 80.05% less degree and 31.49% fewer operations than the seeded predecessor. All variants have thirteen unrestricted integer unknowns and the same original coherent coefficient tuple. The old-source proof is retained in `old_source_section.tex` for an explicit apples-to-apples comparison. No twelve-variable result, variable-count record, global optimum, or reduction of coefficient storage is claimed.

The metric counts every live binary addition, subtraction and multiplication, including scalar multiplication and all fixed-power gates. Reuse and fixed leaves are free. The canonical evaluator uses sparse Horner forms, reuses known exponent sums, and factors the merger cubic as $\rho(\rho^2 - u^2) - u^3$. These are exact polynomial identities, not new semantic assumptions.

The checker independently parses all 22 original residuals, proves the two row inverses and all 22 target polynomial identities, verifies the complete 22,544 isolation interactions, and authenticates the full fixed tuple. It also replays literal circuit registers over three finite fields, checks signed and zero branches, the norm and CRT identities, nonvacuous Pell examples, source positivity, sparse carries, and the parity calibration. These finite implementation audits corroborate the universal proof; they are not a new kernel-checked formal theorem or an unbounded consistency search.

Run `python build_jones.py`, `python check_jones.py`, and `python -O check_jones.py`. The complete division-free polynomial and fixed binding are in the emitted DAG and `FIXED_BINDING.json`. Original source provenance is in `SOURCE_PINS.json`; the CRT source archive and theorem are identified in `CRT_PROVENANCE.json`.

1.8 Recommended compact companion: odd-parity calibration

The recommended compact companion has the same thirteen unrestricted integer unknowns, exact degree 11 543 632, and 518 live binary operations (326 multiplications, 169 additions, 23

subtractions). Its literal DAG is `generated/signed_coprime_compact_split_crt.json`. The 520-operation construction above is retained as the fully explicit baseline. Only three source registers are changed:

$$\mathcal{G}_+ = D_2(B)U_c^2 + D_1(B)U_c + D_0(B) + B^{2161}, \qquad L = B^{2279}, \qquad \widehat{N} = \eta\epsilon B^{1201},$$

with the same $\eta = 2^{8\kappa-10}$. In the literal ratio registers use

$$U = \widehat{N}^2 w^2, \qquad Y = \widehat{N}^2 s.$$

Everything else, including the mask, source targets, interval guard, CRT registers, spectator stack, merger, and raw final norm, is unchanged. The JSON register named `N` contains $\widehat{N}$, which can be negative. *Only in the proof*, instantiate the positive Jones parameter by $N = |\widehat{N}|$. Since all uses of this register in the polynomial are through its square, $N^2 = \widehat{N}^2$ exactly. There is no absolute-value gate or extra unknown in the equation.

Unconditional positivity. The same square completion, now with a slightly lower off-target monomial, gives

$$\begin{aligned}\mathcal{G}_+ &> \tfrac{1}{4}B^{1929}U_c^2 + B^{2161} - 4A_*^2B^{2125} - 2A_*B^{1719}\\ &> \tfrac{1}{4}B^{1929}U_c^2 + \tfrac{1}{2}B^{2161},\end{aligned}$$

because $4A_*^2/B^{36} < 1/4$ and $2A_*/B^{442} < 1/4$. Thus

$$S_c > \tfrac{1}{2}N_0B^{1929}U_c^2 + U_c + N_0B^{2161} - \ell > 0$$

for every integer g, exactly as in the baseline proof. Hence $R > L^2$. For the positive proof-side N one has

$$N^2 = 2^{-21}a^{16}bB^{2402} < a^{15}B^{2403} < B^{2418} < B^{4558} = L^2.$$

Also $N \geq c = bB^{116}$, since $\eta B^{1085} \geq 2|\epsilon|$, using $B \geq 2|\epsilon|$ and $\eta \geq 1$. Finally $N \geq B^{1201} \geq 8$. Thus all hypotheses $R \geq N \geq 8$ and $b, c \leq N$ concern the positive number $|\widehat{N}|$, not signed $\widehat{N}$. This is sufficient for every use of the ratio theorem and for the canonical estimate proving $\Delta > 0$.

A smaller block is sufficient. On the required range $0 < g < c$, the previous bound $2|\mathcal{G}| < B^{2162}/256$ still holds. Also $T_* < B^{2162}/256$ and $2B^{2161} < B^{2162}/128$, since $B > 256$. Therefore

$$0 < J_c < B^{2162}/64, \qquad S_c < B^{2279}/32, \qquad T_c < B^{2279}/64.$$

Consequently $S_c + T_c < L = B^{2279}$. The pad is still above every target, since $2161 > 2129$, and the same coefficient bound $|h_e| < 994A_*b^2 < B/8$ covers its extra coefficient 1. The entire sparse carry proof is therefore unchanged.

The exact square threshold uses the forced odd parity. After ratio soundness establishes $b = 2^k$, the polynomial identity $b = 2\epsilon^2$ with integer $\epsilon \neq 0$ forces k odd and $\epsilon^2 = 2^{k-1}$. Write $\nu = \kappa + 2k$. For $L = B^{2279}$,

$$\begin{aligned}\log_2 L + s_2(T_c) + 1 &= 2418\nu - 31k - 21\\ &= 2418\kappa + 4805k - 21\\ &= 2(8\kappa - 10 + 1201\nu) + (k-1) = \log_2 \widehat{N}^2.\end{aligned}$$

Thus $N^2 = \widehat{N}^2$ has exactly the required central-binomial valuation threshold. No even-k assertion is needed: that branch cannot occur. For completeness choose $\epsilon = 2^r > 0$ and $b = 2^{1+2r}$, which are still cofinal. The canonical ω remains a square. The same $b, c \leq N$ estimate proves the interval guard. All later CRT, nonzero, square-recovery, and sign-bootstrap arguments are identical. This proves the compact companion's equivalence to a source zero and hence to a finite ZFC refutation.

Exact degree and accounting. Here the relevant degrees are

$$\deg a_0 = 261194, \quad \deg a_1 = 2634542, \quad \deg C_* = 4874866, \quad \deg \Sigma = 448475,$$

and $\deg A_0 = 5771816$, $\deg A_1 = 3531492$. Again A_0 dominates X^2 and A_1, and the leading term of the output is A_0^2. Its diagonal coefficient is

$$630137\, a^{2885464} \eta^{960} \pmod{1000003} \neq 0.$$

This proves exact degree 11 543 632, with 518 operations under the same gate metric. The change saves two operations and 2480 degrees against the scaled split baseline, without changing the source tuple or number of unknowns. The independent audit checks the all-integer source minimum, compact block bounds, negative-ϵ normalization, odd-k threshold, and that the signed `N` node feeds only its square.